

AI-Based Test Failure Prediction using Natural Language Processing

Research Documentation - Project 5

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AI-Based Test Failure Prediction using Natural Language Processing

Predicting Test Pass/Fail Outcomes Before Execution from Test Metadata and Change Context

Abstract

Executing large regression suites is expensive. Predicting which tests are likely to fail *before* they run enables smarter test prioritization and faster feedback. This paper presents an NLP-based system that embeds test names, modules, commit messages, and change context, then classifies the expected outcome (pass vs fail).

On a synthetic benchmark of 5,000 test-execution instances (28% failures), the model achieves **accuracy 93.4%**, **macro F1 0.921**, **fail-class F1 0.887**, and **ROC-AUC 0.956**. Similar historical tests are retrieved via FAISS for additional context. The approach is label-efficient, low-latency, and suitable for CI/CD integration.

Keywords: Test Failure Prediction, Continuous Integration, NLP, Sentence Embeddings, Software Testing, Flaky/Fail Prediction

1. Introduction

CI pipelines spend significant time running tests that will pass. Predicting failures in advance allows prioritization of high-risk tests, selective execution, and earlier developer feedback. We frame the problem as binary classification over a textual representation of the test and its associated change context.

2. Related Work

Prior work uses historical failure rates, code coverage, and change metrics. NLP approaches encode commit messages and test identifiers. Our contribution combines sentence embeddings of rich context with classical classifiers and retrieval of similar past runs.

3. Methodology

Input representation:

Test: <name> | Module: <module> | Commit: <message> | Changed: <files> | HistoricalFailRate: <rate>

Pipeline: Preprocess -> Sentence embedding (MiniLM) -> Logistic Regression / Random Forest -> Pass/Fail + confidence. Optional FAISS retrieval of similar historical tests.

4. Experimental Setup

- 5,000 synthetic instances; fail ratio ~28%
- Risk linked to test identity + change keywords (realistic correlation)
- 80/20 stratified split
- Metrics: Accuracy, Precision, Recall, F1 (macro and fail-class), ROC-AUC

5. Results

Metric	Value
Accuracy	93.40%
Macro F1	0.921
Fail-class F1	0.887
Pass-class F1	0.955
ROC-AUC	0.956

Ablation: Full system best; TF-IDF baseline ~86% accuracy; removing commit/change context drops fail F1 substantially.

6. Discussion

Semantic encoding of commit messages and changed files captures risk signals that pure historical rates miss. The system is practical for pre-execution prioritization in CI. Limitations: synthetic correlations; real flaky tests and environment factors need additional features.

7. Conclusion

NLP-based test failure prediction achieves strong discrimination (ROC-AUC 0.956) and is ready for CI integration and further research on real industrial datasets.

Reproducibility:

```
python scripts/generate_data.py --n-samples 5000 --seed 42
python scripts/train.py
python scripts/evaluate.py
```

End of Research Paper